



# Subscription Billing Platform

Engineering take-home assignment

|                       |                           |                              |
|-----------------------|---------------------------|------------------------------|
| Assignment type       | Estimated effort          | Deliverable format           |
| Take-home, individual | Design + build + document | Git repo + docs + demo video |

## Objective

Design and build a subscription product end to end, covering plans, checkout, payments, invoices, notifications, and subscription lifecycle management. We are less interested in visual polish and more interested in your system design decisions, how you handle edge cases, and the overall quality of your implementation.

Build a generic SaaS subscription platform that demonstrates product thinking, payment integration, webhook processing, notification handling, and robust state management.

## Features

- Pricing page with multiple subscription plans.
- Checkout and payment flow, covering success, pending, and failed outcomes.
- Webhook handling with idempotency.
- Subscription lifecycle: active, cancelled, expired.
- Upgrade and downgrade between plans.
- Invoice generation for every successful payment.
- Email notifications for subscription, payment, invoice, and cancellation events.
- Documentation of all architectural decisions in ARCHITECTURE.md.

## Recommended tech stack

| Layer          | Recommended                   | Alternative                             |
|----------------|-------------------------------|-----------------------------------------|
| Frontend       | Next.js 15 + React            | Any modern framework                    |
| Backend        | Node.js                       | Any HTTP framework                      |
| Payments       | Razorpay / Stripe (test mode) | Any gateway                             |
| Notifications  | Resend / AWS SES / SMTP       | Any email provider                      |
| Database       | PostgreSQL                    | Any SQL database (justify if different) |
| Authentication | JWT                           | Session cookies                         |

# Deliverables

- Git repository with clean commit history.
- README.md with setup instructions.
- ARCHITECTURE.md explaining flow decisions and trade-offs.
- 3–5 minute demo video walking through the working product.
- Tests for critical flows, especially webhook idempotency.

## Expected result — reference mockups

The screens below illustrate the expected functional states, not a required visual style. You are free to design your own UI — these are provided so there is no ambiguity about what each feature should actually do.

### 1 Pricing page

Basic

\$9/mo

1 project  
Email support

Select plan

Most popular

Pro

\$29/mo

10 projects  
Priority support  
Analytics

Select plan

Enterprise

\$99/mo

Unlimited projects  
Dedicated manager  
SLA

Select plan

### 2 Checkout — three payment outcomes

✓

Payment successful

Plan activated immediately, invoice generated automatically.

●

Payment pending

Subscription stays inactive until the gateway confirms.

✗

Payment failed

User can retry; no subscription or invoice is created.

### 3 Subscription dashboard — lifecycle, upgrade/downgrade, invoices

Pro plan

Renews on Aug 7, 2026

Active

Upgrade

Downgrade

Cancel subscription

Active

Cancelled

Expired

|             |               |         |          |
|-------------|---------------|---------|----------|
| Jul 7, 2026 | INV-2026-0417 | \$29.00 | Download |
| Jun 7, 2026 | INV-2026-0392 | \$29.00 | Download |

#### 4 Email notifications to trigger

##### Subscription confirmed

Sent right after successful checkout; includes plan name and renewal date.

##### Invoice generated

Sent for every successful payment webhook, linking or attaching the invoice.

##### Payment failed

Sent on decline, with a retry link back to checkout.

##### Cancellation confirmed

Sent on cancel, stating the exact date access ends.